

MUSTAPHA EL HACHMI MAHTI

AI ENGINEER & AGENTIC SYSTEMS ARCHITECT

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PRESENTATION

AI Engineer specialized in developing agentic systems and advanced RAG architectures. Expert in LangChain, LangGraph, and Graph RAG, I design autonomous agents capable of reasoning, planning, and interacting with complex knowledge bases. I transform LLMs into scalable production systems with knowledge graphs (Neo4j), vector databases (Pinecone, pgvector), and multi-agent orchestration to create high-impact AI solutions. Feel free to check my portfolio to explore my projects in detail.

PROFESSIONAL EXPERIENCE

Egobis - AI Knowledge Management Platform, Dataunboxed (Freelance) · [More details](#)

September 2025 — March 2026

Full-stack document management platform powered by Graph RAG and agentic systems.

- **Graph RAG & Knowledge Graphs** : Graph RAG architecture with Neo4j and Graphiti for entity extraction, relationship mapping, and knowledge graph generation from documents
- **AI Agents & LLM Orchestration** : Implementation of autonomous agents with LangChain/LangGraph for intelligent Q&A, automatic classification, and multi-modal semantic search
- **AI Microservices Architecture** : Distributed NestJS + FastAPI stack with Celery workers for asynchronous processing (OCR, embeddings, graph building), PostgreSQL+pgvector for vector search
- **Multi-Platform Applications** : NextJS (React) interface and React Native/Expo mobile app with JWT authentication, Redis rate limiting, and Cloudflare R2 storage
- **LLM Evaluation** : LLM evaluation and tracing with Langfuse to improve RAG quality

Stack: LangChain, LangGraph, Neo4j, FastAPI, NestJS, PostgreSQL+pgvector, Celery, Redis, React (NextJS), React Native

Full Stack Software Engineer, Groupe Actual, Toulouse · [More details](#)

September 2022 — September 2025

- **Web Development** : Design and development of complex React/TypeScript applications (50,000+ users) with optimized user interfaces and sophisticated state management
- **Mobile Applications** : Cross-platform iOS/Android development with React Native/Expo (250,000+ downloads)
- **Backend Architecture** : Design of NestJS microservices with RESTful APIs, PostgreSQL/MongoDB databases
- **Performance & Scalability** : Optimization of web and mobile performance (+30% speed improvement), implementation of caching solutions and real-time monitoring
- **Data Engineering** : BigQuery development for business analytics and development of high-availability data pipelines

Stack: React, TypeScript, React Native, NestJS (NodeJS), MySQL, Redis, GCP

Full Stack Developer (Internship), Groupe Bizness, Toulouse · [More details](#)

May 2022 — August 2022

- **RAG Chatbot** : Built an AI chatbot answering user questions from a vectorized document knowledge base
- **Hybrid Search** : Implemented ChromaDB for vector search and Elasticsearch for keyword-based retrieval
- **Orchestration & Full-Stack** : Orchestrated agents and tools with LangChain, with an Angular frontend and Python FastAPI APIs

Stack: Angular, Python, FastAPI, LangChain, ChromaDB, Elasticsearch

Full Stack Developer, Société JUMP, Toulouse · [More details](#)

November 2019 — June 2021

- **Mission 1 - Storiz LCMS** : Built an internal Slack tool with BoltJS, NestJS, RabbitMQ, and MongoDB to deploy, update, or delete client environments in minutes
- **Automation & Security** : Implemented asynchronous workflows, Bash execution, UUID audit trails, real-time error feedback, and role-based access control
- **Mission 2 - Betty Care** : Developed an Angular/Ionic healthcare platform with NestJS APIs for patient, surgeon, and admin workflows across the surgical journey
- **Real-Time & Mobile** : Integrated WebSockets/SSE, centralized business data in PostgreSQL, and shipped a cross-platform iOS/Android experience

SKILLS

- **AI & Agentic Systems** : LangChain, LangGraph, RAG, Graph RAG, LLM, Autonomous Agents, Prompt Engineering, RL
- **Vector & Knowledge Databases** : Neo4j, Graphiti, Pinecone, pgvector, Elasticsearch, Embeddings, Semantic Search
- **Backend & Microservices** : FastAPI, NestJS, NodeJS, PostgreSQL, MongoDB, Redis, Celery, RabbitMQ, RESTful APIs
- **Full-Stack Development** : TypeScript, Python, React, NextJS, React Native, Docker, CI/CD

EDUCATION

- **Master Software Architect** | EPITECH Engineering School, Toulouse (2023 — 2025)
- **Bachelor DevOps Application Developer & Designer** | EPSI, Toulouse (2022 — 2023) *With distinction from the jury*
- **Associate Web and Mobile Web Developer** | SIMPLON, Toulouse (2021 — 2022)
- **High School Diploma Science and Technology** | IES Salvador Espriu, Girona, Spain (2018 — 2019)

LANGUAGES

Languages: **Italian** (conversational), **English** (fluent), **French** (native), **Arabic** (native), **Catalan** (native), **Spanish** (native)